Lecture Notes on

Light & Distance

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These notes summarize the fifth Astrophysics Club meeting of the 2026-27 year. The question for the meeting: how do we know how far away a star is if we can never go there? We start with light as a measuring stick, then learn how objects in the sky take up angles, and finish by deriving the parallax formula and the parsec. The formula sheet and the parallax diagram we filled in are in the separate group packet.

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1 Light as a Measuring Stick

1.1 The Question

How do we know how far away a star is if we can never go there?

1.2 The Speed of Light

- **Speed.** Light travels at 300,000 km/s, about 7.5 laps around the Earth every second.
- **Distance, speed, time.** Distance equals speed times time:

$$d = ct.$$

- **The light-year** is the distance light travels in one year:

$$1 \text{ ly} = (3.0 \times 10^8 \text{ m/s})(3.16 \times 10^7 \text{ s}) \approx 9.5 \times 10^{15} \text{ m}.$$

1.3 Mirrors on the Moon

Apollo astronauts left mirrors on the Moon. The slide shows the Laser Ranging Retroreflector the Apollo 11 crew set up at Tranquility Base in 1969.

1.4 Lunar Laser Ranging

We fire a laser at the mirrors and time how long the light takes to come back.

- **Measured round trip.** $t = 2.56$ s.
- **Divide by two.** The light goes there and back, so it covers the distance twice:

$$d = \frac{ct}{2} = \frac{(3.0 \times 10^8 \text{ m/s})(2.56 \text{ s})}{2} \approx 3.84 \times 10^8 \text{ m}.$$

- **Check.** The Earth-Moon distance is 384,400 km.

1.5 Looking Back in Time

Light takes time to reach us, so we always see objects as they were when the light left them.

- **The Sun.** 8.3 minutes.
- **Proxima Centauri.** 4.2 years.
- **Betelgeuse.** 550 years.

If Betelgeuse exploded tonight, we would not see it for about 550 years. It is expected to explode in about 100,000 years.

2 Angles in the Sky

2.1 Angle Conversions

- $180^\circ = \pi$ rad.
- $1^\circ = 60'$ (arcminutes).
- $1' = 60''$ (arcseconds).

2.2 Objects in the Sky Take Up Angles

- **The full Moon.** About 0.5° across.
- **The Sun.** About 0.5° across.
- **The Andromeda galaxy.** About 3° across.

2.3 The Small-Angle Approximation

- **Arc length.** On a circle of radius r, an angle θ cuts off an arc

$$a = r\theta, \qquad \text{so} \qquad \theta = \frac{a}{r} \quad \text{(radians)}.$$

- **Far away.** When the object is far away, the arc a is almost the object's size D, and the radius r is almost the distance d:

$$a \approx D, \qquad r \approx d, \qquad \text{so} \qquad \theta \approx \frac{D}{d}.$$

2.4 Example: The Sun

How big does the Sun look from Earth?

- **Size.** $D = 2(6.96 \times 10^8 \text{ m}) = 1.392 \times 10^9 \text{ m}$.
- **Distance.** $d = 1.50 \times 10^{11}$ m.
- **Angle.**

$$\theta \approx \frac{D}{d} = \frac{1.392 \times 10^9 \text{ m}}{1.50 \times 10^{11} \text{ m}} = 9.28 \times 10^{-3} \text{ rad}.$$

- **In arcminutes.**

$$(9.28 \times 10^{-3} \text{ rad})(3438'/\text{rad}) \approx 32',$$

about half a degree.

2.5 Fun Fact

The Sun ($32'$) and the Moon ($31'$ on average) look almost the same size in our sky. The slide shows the total solar eclipse of August 21, 2017, seen from Madras, Oregon.

3 Parallax

3.1 Parallax With Your Thumb

1. Hold your thumb out at arm's length.
2. Close one eye, then the other.
3. Watch your thumb against the background.

Your thumb seems to jump against the background when you switch eyes.

3.2 Deriving the Parallax Formula

Earth goes around the Sun, so six months apart (January and July) we see a nearby star from opposite sides of the Sun. The nearby star seems to shift against the distant stars behind it.

- **The triangle.** The Sun, the Earth, and the star make a right triangle. The side from the Sun to the Earth is 1 AU, the side from the Sun to the star is d, and the angle at the star is the parallax angle p.
- **Tangent.**
$$\tan p = \frac{1\text{ AU}}{d}.$$
- **Tiny angle.** p is tiny, so we use the small-angle approximation, $\tan p \approx p$:
$$p \approx \frac{1\text{ AU}}{d} \quad (p \text{ in radians}).$$

3.3 The Parsec

- **Switch to arcseconds.** 1 rad = 206,265″, so
$$d = \frac{206{,}265\text{ AU}}{p\,[''] }.$$
- **Define the parsec** as that distance: 1 pc = 206,265 AU. Then
$$d = \frac{1}{p}, \qquad d \text{ in parsecs}, \quad p \text{ in arcseconds}.$$
- **Size of a parsec.**
$$1\text{ pc} = 206{,}265\text{ AU} = 3.09 \times 10^{16}\text{ m} = 3.26\text{ light-years}.$$

3.4 Example: Proxima Centauri

- **Parallax.** $p = 0.768''$.
- **Distance.**
$$d = \frac{1}{0.768}\text{ pc} = 1.30\text{ pc}.$$
- **In light-years.**
$$(1.30\text{ pc})(3.26\text{ ly/pc}) \approx 4.2\text{ light-years}.$$

This matches the 4.2 years Proxima Centauri's light takes to reach us.

3.5 Distance-Chasing at Scale

ESA's Gaia mission has measured the parallax of 1.47 billion objects (2022), almost all of them stars.

4 Team Kahoot

We finished with a team Kahoot: 4 questions, 1 minute per question, with the formula sheet and the group packet or scratch paper.